

SAMUEL KOSASIH

AI ENGINEER

PROFESSIONAL SUMMARY

Innovative and analytical AI and Data Engineering student at Nanyang Polytechnic (graduating in April 2026) with a strong foundation in machine learning, data analysis, and software development. Experienced in designing automated systems, dashboard development, and predictive analytics. Adept at working with Python, SQL, TensorFlow, and Power BI to deliver impactful insights. Passionate about leveraging technology to enhance efficiency and solve real-world problems.

WORK & PROJECT EXPERIENCE

Automated Defect Detection System (EGT217 AI Engineering Project)

- Developed an AI-powered defect detection system using computer vision for Watson EP, an ODM/OEM company in Singapore.
- Implemented data augmentation and labeling techniques to enhance model accuracy.
- Utilized TensorFlow and PyTorch for machine learning model training and optimization.
- Improved defect detection efficiency, reducing manual inspection time by 40%.

Real-time Air Quality & Machine Condition Monitoring (Plastic Technology Lab)

- Designed and implemented an IoT-based air quality monitoring system using Arduino R4 WiFi.
- Integrated sensors for temperature, humidity, and air quality, transmitting data via MQTT to MySQL.
- Processed and visualized data using Python and Power BI for real-time insights.
- Overcame hardware limitations and optimized data collection strategies for better efficiency.

Predicting Rain with Raspberry Pi

- Deployed Raspberry Pi 4 devices across four locations to collect environmental data.
- Captured sky images every 5 minutes and extracted cloud-related metrics.
- Developed a machine learning model to predict rainfall based on sensor and image data.
- Achieved significant accuracy improvements, contributing to early weather prediction efforts.

CONTACT

+65 8899 7938

SamuelKoh5104@gmail.com

<https://www.linkedin.com/in/samuel-kosasih-44894a253/>

EDUCATION

Nanyang Polytechnic (NYP)

Diploma in AI & Data Engineering (April 2026)

KEY SKILLS

- Programming & Software Development: Python, Java, SQL, HTML, CSS, C (Arduino)
- Machine Learning & Data Science: TensorFlow, PyTorch, SMOTE, Gradient Boosting, Random Forest, Naïve Bayes, PCA, Clustering
- Data Engineering & Analysis: Google Colab, Power BI, MySQL, Data Cleaning, Dashboard Development
- Hardware & Automation: Raspberry Pi 4, Arduino R4 WiFi, MQTT, FPWIN Pro7, Ladder Programming
- Cloud & Web Applications: AWS EC2, RDS, S3, Auto Scaling, ALB, VPC, Lambda Functions

SOFT SKILLS

- Teamwork
- Problem-Solving
- Attention to Detail
- Communication
- Adaptability

CERTIFICATIONS

- Certificate in Database Design & Administration (Oracle Academy)
- Certificate in Predictive Analytics in Data Science (SAS)
- AI-900 Microsoft Azure AI Fundamentals